

CHAPTER 16

one-factor interest rate modeling

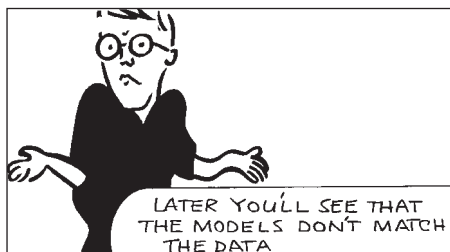


The aim of this Chapter...

... is to model interest rates as random walks and bring together the instruments of the fixed-income world and the modeling ideas of Black and Scholes. You will see many familiar ideas and a few new ones that are not seen in the context of equity derivatives.

In this Chapter...

- stochastic models for interest rates
- how to derive the bond pricing equation for many fixed-income products
- the structure of many popular interest rate models



16.1 INTRODUCTION

Until now I have assumed that interest rates are either constant or a known function of time. This may be a reasonable assumption for short-dated equity contracts. But for longer-dated contracts the interest rate must be more accurately modeled. This is not an easy task. In this chapter I introduce the ideas behind modeling interest rates using a single source of

randomness. This is **one-factor interest rate modeling**. The model will allow the short-term interest rate, the spot rate, to follow a random walk. This model leads to a parabolic partial differential equation for the prices of bonds and other interest rate derivative products.

The 'spot rate' that we will be modeling is a very loosely defined quantity, meant to represent the yield on a bond of infinitesimal maturity. In practice one should take this rate to be the yield on a liquid finite-maturity bond, say one of one month. Bonds with one day to expiry do exist but their price is not necessarily a guide to other short-term rates. I will continue to be vague about the precise definition of the spot interest rate. We could argue that if we are pricing a complex product that is highly model dependent then the *exact* definition of the independent variable will be relatively unimportant compared with the choice of model.

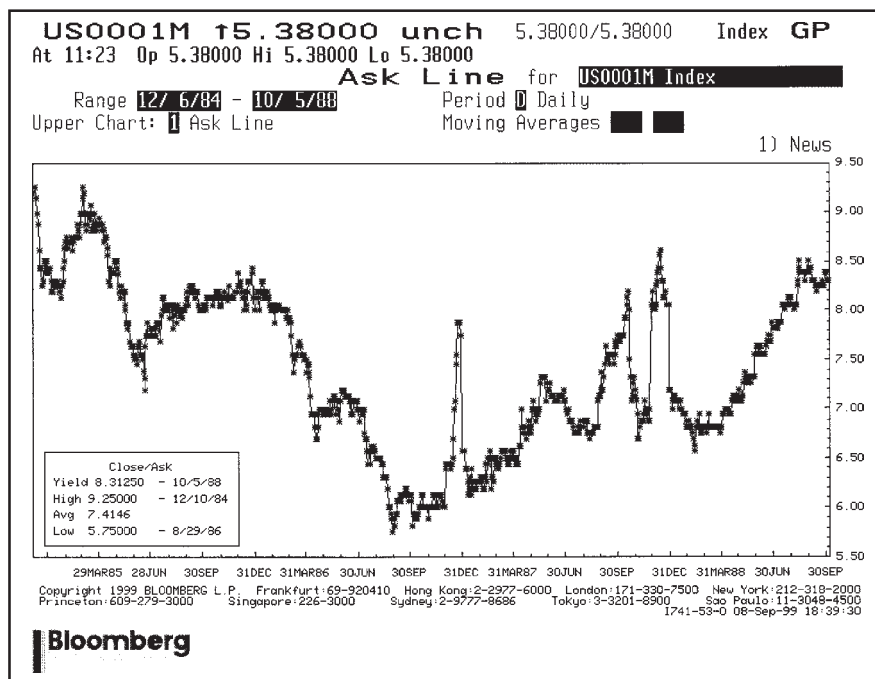


Figure 16.1 One-month interest rate time series. Source: Bloomberg L.P.

16.2 STOCHASTIC INTEREST RATES

Since we cannot realistically forecast the future course of an interest rate, it is natural to model it as a random variable. We are going to model the behavior of r , the interest rate received by the shortest possible deposit. From this we will see the development of a model for all other rates. The interest rate for the shortest possible deposit is commonly called the **spot interest rate**.

Figure 16.1 shows the time series of a one-month US interest rate. We will often use the one-month rate as a proxy for the spot rate.

Earlier I proposed a model for the asset price as a stochastic differential equation, the lognormal random walk. Now let us suppose that the interest rate r is governed by another stochastic differential equation of the form

$$dr = u(r, t)dt + w(r, t)dX. \quad (16.1)$$

The functional forms of $u(r, t)$ and $w(r, t)$ determine the behavior of the spot rate r . For the present I will not specify any particular choices for these functions. We use this random walk to derive a partial differential equation for the price of a bond using similar arguments to those in the derivation of the Black–Scholes equation. Later I describe functional forms for u and w that have become popular with practitioners.

Time Out...

Intuition behind stochastic interest rates

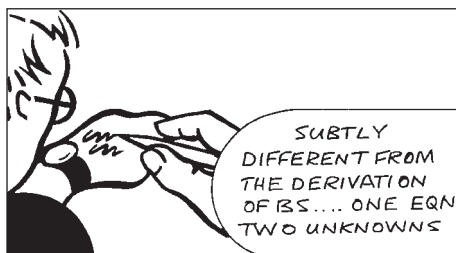
Equation (16.1) is just another recipe for generating random numbers. Until now we've concentrated on the lognormal random walk as the model for asset prices. But there's no reason why interest rates should behave like stock prices, there's no reason why we should use the same model for interest rates as for equities. In fact, such a model would be a very poor one; interest rates certainly do not exhibit the long-term exponential growth seen in the equity markets.

So, we need another model. But we're going to use the same mathematical, stochastic framework, with subtly and suitably different forms. Modeling interest rates in this framework amounts to choosing functional forms for the dt and dX coefficients in our random walk recipe.

From a model for the short-term interest rate r will follow a model for bonds of all maturities and hence interest rates for all maturities. In other words, the spot interest rate model leads to a model for the whole forward curve.

I'll be taking the stochastic calculus and differential equation approach to the pricing of interest rate products. But it can all be done in a binomial or trinomial setting. Actually, trinomial is the more popular for interest rate products. The principle is the same as in the equity tree model. I'll give some details shortly.





16.3 THE BOND PRICING EQUATION FOR THE GENERAL MODEL

When interest rates are stochastic a bond has a price of the form $V(r, t; T)$. The reader should think for the moment in terms of simple bonds, but the governing equation will be far more general and may be used to price many other contracts. That's why I'm using the

notation V rather than our earlier Z for zero-coupon bonds.

Pricing a bond presents new technical problems, and is in a sense harder than pricing an option since *there is no underlying asset with which to hedge*. We are therefore not modeling a *traded* asset; the traded asset (the bond, say) is a derivative of our independent variable r . The only way to construct a hedged portfolio is by hedging one bond with a bond of a different maturity. We set up a portfolio containing two bonds with different maturities T_1 and T_2 . The bond with maturity T_1 has price $V_1(r, t; T_1)$ and the bond with maturity T_2 has price $V_2(r, t; T_2)$. We hold one of the former and a number $-\Delta$ of the latter. We have

$$\Pi = V_1 - \Delta V_2. \quad (16.2)$$

The change in this portfolio in a time dt is given by

$$d\Pi = \frac{\partial V_1}{\partial t} dt + \frac{\partial V_1}{\partial r} dr + \frac{1}{2} w^2 \frac{\partial^2 V_1}{\partial r^2} dt - \Delta \left(\frac{\partial V_2}{\partial t} dt + \frac{\partial V_2}{\partial r} dr + \frac{1}{2} w^2 \frac{\partial^2 V_2}{\partial r^2} dt \right), \quad (16.3)$$

where we have applied Itô's lemma to functions of r and t . Which of these terms are random? Once you've identified them you'll see that the choice

$$\Delta = \frac{\partial V_1}{\partial r} \bigg/ \frac{\partial V_2}{\partial r}$$

eliminates all randomness in $d\Pi$. This is because it makes the coefficient of dr zero. We then have

$$\begin{aligned} d\Pi &= \left(\frac{\partial V_1}{\partial t} + \frac{1}{2} w^2 \frac{\partial^2 V_1}{\partial r^2} - \left(\frac{\partial V_1}{\partial r} \bigg/ \frac{\partial V_2}{\partial r} \right) \left(\frac{\partial V_2}{\partial t} + \frac{1}{2} w^2 \frac{\partial^2 V_2}{\partial r^2} \right) \right) dt \\ &= r \Pi dt = r \left(V_1 - \left(\frac{\partial V_1}{\partial r} \bigg/ \frac{\partial V_2}{\partial r} \right) V_2 \right) dt, \end{aligned}$$

where we have used arbitrage arguments to set the return on the portfolio equal to the risk-free rate. This risk-free rate is just the spot rate.

Collecting all V_1 terms on the left-hand side and all V_2 terms on the right-hand side we find that

$$\frac{\frac{\partial V_1}{\partial t} + \frac{1}{2} w^2 \frac{\partial^2 V_1}{\partial r^2} - r V_1}{\frac{\partial V_1}{\partial r}} = \frac{\frac{\partial V_2}{\partial t} + \frac{1}{2} w^2 \frac{\partial^2 V_2}{\partial r^2} - r V_2}{\frac{\partial V_2}{\partial r}}.$$

At this point the distinction between the equity and interest rate worlds starts to become apparent. This is *one* equation in *two* unknowns. Fortunately, the left-hand side is a

function of T_1 but not T_2 and the right-hand side is a function of T_2 but not T_1 . The only way for this to be possible is for both sides to be independent of the maturity date. Dropping the subscript from V , we have

$$\frac{\frac{\partial V}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} - rV}{\frac{\partial V}{\partial r}} = a(r, t)$$

for some function $a(r, t)$. I shall find it convenient to write

$$a(r, t) = w(r, t)\lambda(r, t) - u(r, t);$$

for a given $u(r, t)$ and non-zero $w(r, t)$ this is always possible. The function $\lambda(r, t)$ is as yet unspecified. λ is 'universal' in that through this function all interest rate products are linked.

The bond pricing equation is therefore

$$\frac{\partial V}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda w) \frac{\partial V}{\partial r} - rV = 0. \quad (16.4)$$

Time Out...

Is this like Black-Scholes?

Pretty much, yes. Mathematically, it's of the same form as the Black-Scholes equation, but with different coefficients in front of two of the partial derivative terms. That's why I like to teach people about BS before interest rates... the math is almost identical but there are no problems with one equation for two unknowns.

The downside of this kind of modeling for interest rates is rather severe. Finding the best (correct?) form for w and $u - \lambda w$ is not easy. And it's not even possible to determine $u - \lambda w$ from observing time series for r , since that time series depends on u and w not on λ .



To find a unique solution of (16.4) we must impose one final and two boundary conditions. The final condition corresponds to the payoff on maturity and so for a zero-coupon bond

$$V(r, T; T) = 1.$$

Boundary conditions depend on the form of $u(r, t)$ and $w(r, t)$ and are discussed later for a special model.

It is easy to incorporate coupon payments into the model. If an amount $K(r, t) dt$ is received in a period dt then

$$\frac{\partial V}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda w) \frac{\partial V}{\partial r} - rV + K(r, t) = 0.$$

When this coupon is paid discretely, arbitrage considerations lead to jump condition

$$V(r, t_c^-; T) = V(r, t_c^+; T) + K(r, t_c),$$

where a coupon of $K(r, t_c)$ is received at time t_c .



Time Out...

Pricing by binomial and trinomial trees

Remember how we built up the binomial tree in Chapter 3 for equities? The process is the same for interest rate products, after all, the pricing differential equation is mathematically very similar to the Black–Scholes equation.

Here's how the binomial model works. There are several stages.

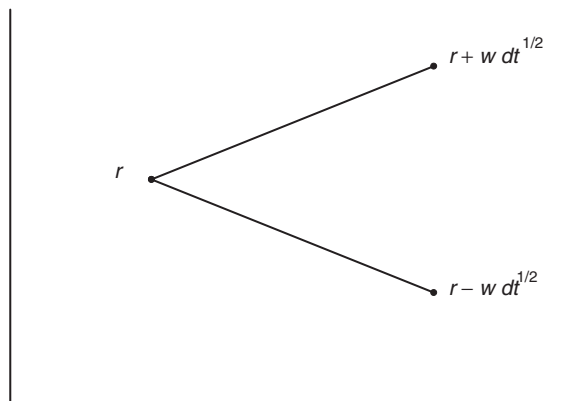
Stage 1: Build your tree There are several possibilities for this, just as there were when building up the equity tree. The simplest is to put all the diffusion into the up and down moves. For example, the interest rate r goes to

$$r + w \delta t^{1/2}$$

on an up move, or

$$r - w \delta t^{1/2}$$

on a down move. See the figure below.



Stage 2: Define the risk-neutral probabilities Simple. The probability of an up move is

$$\frac{1}{2} + \frac{u \delta t^{1/2}}{2w}.$$

But the *risk-neutral* probability is

$$\frac{1}{2} + \frac{(u - \lambda w) \delta t^{1/2}}{2w}.$$

It's the risk-neutral probability you will use when working out expected values.

Stage 3: Discounting Discount at the rate r at the base of the two branches.

Now you just follow the same procedure as in Chapter 3 to work out contract values. You could even modify the VB code in that chapter for interest rate products.

Often trinomial models are used because of the extra degree of freedom they allow in choosing parameters—you are still only going to fit the volatility and risk-neutral drift.

16.4 WHAT IS THE MARKET PRICE OF RISK?

I now give an interpretation of the function $\lambda(r, t)$. Imagine that you hold an unhedged position in one bond with maturity date T . In a time-step dt this bond changes in value by

$$dV = w \frac{\partial V}{\partial r} dX + \left(\frac{\partial V}{\partial t} + \frac{1}{2} w^2 \frac{\partial^2 V}{\partial r^2} + u \frac{\partial V}{\partial r} \right) dt.$$

From (16.4) this may be written as

$$dV = w \frac{\partial V}{\partial r} dX + \left(w \lambda \frac{\partial V}{\partial r} + rV \right) dt,$$

or

$$dV - rV dt = w \frac{\partial V}{\partial r} (dX + \lambda dt). \quad (16.5)$$

The right-hand side of this expression contains two terms: a deterministic term in dt and a random term in dX . The presence of dX in (16.5) shows that this is not a riskless portfolio. The deterministic term may be interpreted as the excess return above the risk-free rate for accepting a certain level of risk. In return for taking the extra risk the portfolio profits by an extra λdt per unit of extra risk, dX . The function λ is therefore called the **market price of risk**.



16.5 INTERPRETING THE MARKET PRICE OF RISK, AND RISK NEUTRALITY

The bond pricing equation (16.4) contains references to the functions $u - \lambda w$ and w . The former is the coefficient of the first-order derivative with respect to the spot rate, and the latter appears in the coefficient of the diffusive, second-order derivative. The four terms in the equation represent, in order as written, time decay, diffusion, drift and discounting. We can interpret the solution of this bond pricing equation as the expected present value of all cashflows, just like we could with equity derivatives.

Suppose that we get a 'Payoff' at time T then the value of that contract today would be

$$E \left[e^{-\int_t^T r(\tau) d\tau} \text{Payoff} \right].$$

Notice that the present value (exponential) term goes inside the expectation since it is also random when interest rates are random.

We exploit this relationship in Chapter 29, and see exactly how to price via simulations.

As with equity options, this expectation is not with respect to the *real* random variable, but instead with respect to the *risk-neutral* variable. There is this difference because the drift term in the equation is not the drift of the real spot rate u , but the drift of another rate, called the **risk-neutral spot rate**. This rate has a drift of $u - \lambda w$. When pricing interest rate derivatives (including bonds of finite maturity) it is important to model, and price, using the risk-neutral rate. This rate satisfies

$$dr = (u - \lambda w) dt + w dX.$$

We need the new market-price-of-risk term because our modeled variable, r , is not traded.

Because we can't observe the function λ , except possibly via the whole yield curve (see Chapter 17), I tend to think of it as a great big carpet under which we can brush all kinds of nasty, inconvenient things.



16.6 NAMED MODELS

There are many interest rate models associated with the names of their inventors. The stochastic differential equation (16.1) for the risk-neutral interest rate process incorporates the models of Vasicek, Cox, Ingersoll & Ross, Ho & Lee, and Hull & White.

16.6.1 Vasicek

The Vasicek model takes the form

$$dr = (\eta - \gamma r)dt + \beta^{1/2}dX.$$

This model is so 'tractable' that there are explicit formulæ for many interest rate derivatives. The value of a zero-coupon bond is given by

$$e^{A(t;T) - rB(t;T)}$$

where

$$B = \frac{1}{\gamma}(1 - e^{-\gamma(T-t)})$$

and

$$A = \frac{1}{\gamma^2}(B(t; T) - T + t)(\eta\gamma - \frac{1}{2}\beta) - \frac{\beta B(t; T)^2}{4\gamma}.$$

The model is mean reverting to a constant level, which is a good property, but interest rates can easily go negative, which is a very bad property.

In Figure 16.2 are shown three types of yield curves predicted by the Vasicek model, each uses different parameters. (It is quite difficult to get the humped yield curve with reasonable numbers.)

16.6.2 Cox, Ingersoll & Ross

The CIR model takes the form

$$dr = (\eta - \gamma r)dt + \sqrt{\alpha r} dX.$$

The spot rate is mean reverting and if $\eta > \alpha/2$ the spot rate stays positive. There are some explicit solutions for interest rate derivatives, although typically involving integrals of the non-central chi-squared distribution. The



Vasicek
simulations and
yield curves

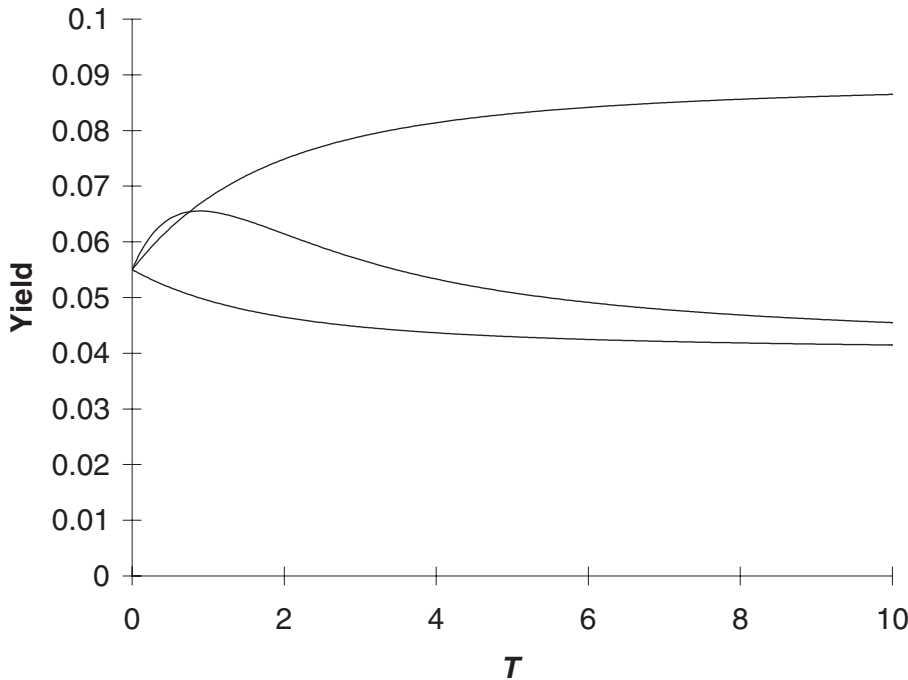


Figure 16.2 Three types of yield curve given by the Vasicek model.

value of a zero-coupon bond is

$$e^{A(t;T)-rB(t;T)}$$

where A and B are given by

$$\frac{\alpha}{2}A = a\psi_2 \log(a - B) + \psi_2 b \log((B + b)/b) - a\psi_2 \log a,$$

and

$$B(t; T) = \frac{2(e^{\psi_1(T-t)} - 1)}{(\gamma + \psi_1)(e^{\psi_1(T-t)} - 1) + 2\psi_1}$$

where

$$\psi_1 = \sqrt{\gamma^2 + 2\alpha} \quad \text{and} \quad \psi_2 = \frac{\eta}{a + b}$$

and

$$b, a = \frac{\pm\gamma + \sqrt{\gamma^2 + 2\alpha}}{\alpha}.$$

In Figure 16.3 are simulations of the Vasicek and CIR models using the same random numbers. The parameters have been chosen to give similar mean and standard deviations for the two processes.



CIR simulations
and yield curves

16.6.3 Ho & Lee

Ho & Lee takes the form

$$dr = \eta(t) dt + \beta^{1/2} dX.$$

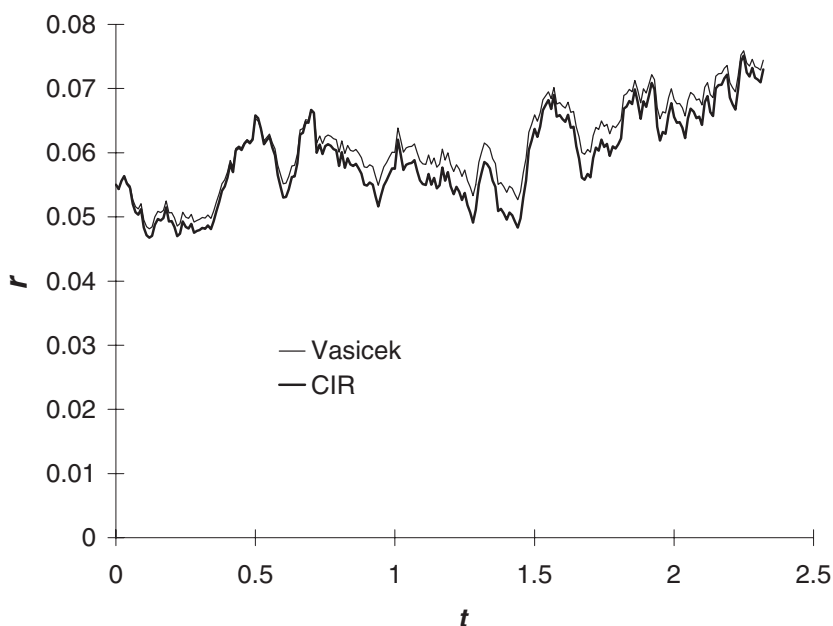


Figure 16.3 A simulation of the Vasicek and CIR models using the same random numbers.

Note the function of time in this. The value of zero-coupon bonds is given by

$$e^{A(t;T) - rB(t;T)}$$

where

$$B = T - t$$

and

$$A = - \int_t^T \eta(s)(T - s) ds + \frac{1}{6} \beta (T - t)^3.$$

This model was the first ‘no-arbitrage model’ of the term structure of interest rates. By this is meant that the careful choice of the function $\eta(t)$ will result in theoretical zero-coupon bonds prices, output by the model, which are the same as market prices. This technique is also called **yield curve fitting**. This careful choice is

$$\eta(t) = - \frac{\partial^2}{\partial t^2} \log Z_M(t^*; t) + \beta(t - t^*)$$

where today is time $t = t^*$. In this $Z_M(t^*; T)$ is the market price today of zero-coupon bonds with maturity T . Clearly this assumes that there are bonds of all maturities and that the prices are twice differentiable with respect to the maturity. We’ll see the mathematics of this in the next chapter.

16.6.4 Hull & White

Hull & White have extended both the Vasicek and the CIR models to incorporate time-dependent parameters. This time dependence again allows the yield curve (and even a volatility structure) to be fitted.

16.7 EQUITY AND FX FORWARDS AND FUTURES WHEN RATES ARE STOCHASTIC

Recall from Chapter 6 that forward prices and futures prices are the same if rates are constant. How does this change, if at all, when rates are stochastic? We must repeat the analysis of that chapter but now with

$$dS = \mu S dt + \sigma S dX_1$$

and

$$dr = u(r, t) dt + w(r, t) dX_2.$$

We are in the world of correlated random walks, as described in Chapter 12. The correlation coefficient is ρ .

16.7.1 Forward contracts

$V(S, r, t)$ will be the value of the forward contract at any time during its life on the underlying asset S , and maturing at time T . As in Chapter 6, I’ll assume that the delivery price is known and then find the forward contract’s value.

Set up the portfolio of one long forward contract and short Δ of the underlying asset, and Δ_1 of a risk-free bond:

$$\Pi = V(S, t) - \Delta S - \Delta_1 Z.$$

I won't go through all the details, because the conclusion is the obvious one:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \rho\sigma S w \frac{\partial^2 V}{\partial S \partial r} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} + rS \frac{\partial V}{\partial S} + (u - \lambda w) \frac{\partial V}{\partial r} - rV = 0.$$

The final condition for the equation is simply the difference between the asset price S and the fixed delivery price $\bar{S}$. So

$$V(S, r, T) = S - \bar{S}.$$

The solution of the equation with this final condition is

$$V(S, r, t) = S - \bar{S}Z.$$

At this point Z is not just any old risk-free bond, it is a zero-coupon bond having the same maturity as the forward contract. This is the forward contract's value during its life.

Remember that the delivery price is set initially $t = t_0$ as the price that gives the forward contract zero value. If the underlying asset is S_0 at t_0 then

$$0 = S_0 - \bar{S}Z$$

or

$$\bar{S} = \frac{S_0}{Z}.$$

The quoted forward price is therefore

$$\text{Forward price} = \frac{S}{Z}.$$

Remember that Z satisfies

$$\frac{\partial Z}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 Z}{\partial r^2} + (u - \lambda w) \frac{\partial Z}{\partial r} - rZ = 0$$

with

$$Z(r, T) = 1.$$

16.8 FUTURES CONTRACTS

Use $F(S, r, t)$ to denote the futures price.

Set up a portfolio of one long futures contract and short Δ of the underlying, and Δ_1 of a risk-free bond:

$$\Pi = -\Delta S - \Delta_1 Z.$$

(Remember that the futures contract has no value.)

$$d\Pi = dF - \Delta dS - \Delta_1 dZ.$$

Following the usual routine we get

$$\frac{\partial F}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} + \rho\sigma Sw \frac{\partial^2 F}{\partial S \partial r} + \frac{1}{2}w^2 \frac{\partial^2 F}{\partial r^2} + rS \frac{\partial F}{\partial S} + (u - \lambda w) \frac{\partial F}{\partial r} = 0.$$

The final condition is

$$F(S, r, T) = S.$$

Let's write the solution of this as

$$F(S, r, t) = \frac{S}{p(r, t)}.$$

Why? Two reasons. First, a similarity solution is to be expected, the price should be proportional to the asset price. Second, I want to make a comparison between the futures price and the forward price. The latter is

$$\frac{S}{\bar{Z}}.$$

So it's natural to ask, how similar are Z and p ?

It turns out that p satisfies

$$\frac{\partial p}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 p}{\partial r^2} + (u - \lambda w) \frac{\partial p}{\partial r} - \underbrace{rp - w^2 \frac{\left(\frac{\partial p}{\partial r}\right)^2}{q}} + \rho\sigma\beta \frac{\partial p}{\partial r} = 0. \quad (16.6)$$

(Just plug the similarity form into the equation to see this.)

The final condition is

$$p(r, T) = 1.$$

The differences between the p and Z equations are in the underlined terms in Equation (16.6).

16.8.1 The convexity adjustment

There is clearly a difference between the prices of forwards and futures when interest rates are stochastic. From Equation (16.6) you can see that the difference depends on the volatility of the spot interest rate, the volatility of the underlying and the correlation between them. Provided that $\rho \geq 0$ the futures price is always greater than the equivalent forward price. Should the correlation be zero then the volatility of the stock is irrelevant. If the interest rate volatility is zero then rates are deterministic and forward and futures prices are the same.

Since the difference in price between forwards and futures depends on the spot rate volatility, market practitioners tend to think in terms of **convexity adjustments** to get from one to the other. Clearly, the convexity adjustment will depend on the precise nature of the model. For the popular models, described above, the p equation (16.6) still has simple solutions.

16.9 SUMMARY

In this chapter I introduced the idea of a random interest rate. The interest rate that we modeled was the ‘spot rate,’ a short-term interest rate. Several popular spot rate models were described. These models were chosen because simple forms of the coefficients make the solution of the basic bond pricing equation straightforward analytically.

FURTHER READING

- See the original interest rate models by Vasicek (1977), Dothan (1978), Cox *et al.* (1985), Ho & Lee (1986) and Black *et al.* (1990).
- For details of the general affine model see the papers by Pearson & Sun (1989), Duffie (1992), Klugman (1992) and Klugman & Wilmott (1994).
- The comprehensive book by Rebonato (1996) describes all of the popular interest rate models in detail.
- Multi-factor models and some very new ideas are discussed in *PWOQF2*.

EXERCISES

1. Substitute

$$Z(r, t; T) = e^{A(t; T) - rB(t; T)},$$

into the bond pricing equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda w) \frac{\partial V}{\partial r} - rV = 0.$$

What are the explicit dependencies of the functions in the resulting equation?

2. Simulate random walks for the interest rate to compare the different named models suggested in this chapter.
3. What final condition (payoff) should be applied to the bond pricing equation for a swap, cap, floor, zero-coupon bond, coupon bond and a bond option?
4. What form does the bond pricing equation take when the interest rate satisfies the Vasicek model

$$dr = (\eta - \gamma r)dt + \beta^{1/2}dX?$$

Solve the resulting equations for A and B in this case, to find

$$A = \frac{1}{\gamma^2}(B - T + t)(\eta\gamma - \frac{1}{2}\beta) - \frac{\beta B^2}{4\gamma},$$

and

$$B = \frac{1}{\gamma}(1 - e^{-\gamma(T-t)}).$$